

# Gaussian Differential Privacy Curve

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## Definition

For  $\mu \geq 0$ , the Gaussian differential privacy tradeoff curve is

$$\beta_{\mu}(\alpha) = \Phi\left(\Phi^{-1}(1 - \alpha) - \mu\right).$$

Its privacy profile is

$$\delta_{\mu}(\epsilon) = \Phi\left(-\frac{\epsilon}{\mu} + \frac{\mu}{2}\right) - e^{\epsilon}\Phi\left(-\frac{\epsilon}{\mu} - \frac{\mu}{2}\right),$$

with the value at  $\mu = 0$  defined by continuity as zero.

## Composition

Independent Gaussian privacy losses compose by addition of their squared parameters. Thus, for curves with parameters  $\mu_i$ ,

$$\mu_{\text{comp}} = \sqrt{\sum_i \mu_i^2}.$$

The implementation encloses each arithmetic operation with directed interval arithmetic and rejects a non-finite resulting parameter.